Divya Bharti

B.Tech Student in AI and Data Science | Aspiring AI/ML Engineer Intern

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Summary

Third-year B.Tech student in AI and Data Science with hands-on project experience in machine learning, deep learning, NLP, computer vision and LLM-based applications. Builds end-to-end AI systems: model training and evaluation, FastAPI backends, and React interfaces. Focused on reliable, testable, production-oriented AI engineering.

Projects

Semantic Cache RAG Engine — FastAPI, React, TypeScript, Qdrant, Sentence Transformers · GitHub

- Built a full-stack RAG application that ingests .txt and PDF files, chunks and embeds them (all-MiniLM-L6-v2), and indexes them in Qdrant to return answers with citations and retrieval scores.
- Implemented a tenant-scoped semantic answer cache (similarity threshold, TTL, invalidation on corpus change) and confidence-based escalation across local and optional LLM providers (OpenAI, Groq, Gemini, Ollama); wrote isolated pytest tests with mocked providers.

Language Translation AI — PyTorch, Hugging Face Transformers (MarianMT), FastAPI, React, Docker · GitHub

- Built a translation web app that runs MarianMT neural models locally for English to and from French, German, Spanish and Hindi, with no third-party translation API.
- Added in-memory caching, input validation, latency logging, automatic CPU/GPU selection, a mocked pytest suite and a Dockerized FastAPI backend.

Mental Health Score Predictor — Scikit-learn, Pandas, FastAPI · GitHub

- Compared Linear Regression and Random Forest on a 5,000-row student dataset using a preprocessing pipeline (scaling, log transform, ordinal and one-hot encoding); Random Forest scored test R^2 0.878 vs 0.740 for Linear Regression on a single 70/30

split, with a noted train-test gap.
- Served the saved pipeline through a FastAPI prediction endpoint and web form (educational, non-clinical demo).

Face Emotion Detection — TensorFlow, Keras, OpenCV · GitHub

- Built a CNN (three convolutional blocks, dropout, Adam) to classify 7 emotions on a ~49,800-image dataset, with augmentation, early stopping and a confusion-matrix and classification-report evaluation script.
- Integrated OpenCV Haar Cascade face detection for single-image and real-time webcam inference.

Technical Skills

- **Languages:** Python, SQL, JavaScript, TypeScript, Java (basics), R
- **Data and Math:** NumPy, Pandas, Matplotlib, Seaborn, EDA, Feature Engineering, Linear Algebra, Probability, Statistics, Gradient Descent
- **Machine Learning:** Scikit-learn, Linear/Logistic Regression, Decision Trees, Random Forest, KNN, SVM, K-Means, XGBoost, Cross-Validation, Hyperparameter Tuning, Regularization; evaluation with Accuracy, Precision, Recall, F1, Confusion Matrix, ROC-AUC, MAE, RMSE, R^2
- **Deep Learning and CV:** PyTorch, TensorFlow, Keras, ANN, CNN, RNN, LSTM, Transformers, Attention, Transfer Learning, OpenCV, YOLO
- **NLP and Generative AI:** Hugging Face Transformers, Text Preprocessing, TF-IDF, Machine Translation, LLMs, Prompt Engineering, Embeddings, Semantic Search, RAG, Qdrant, LangChain, LangGraph, AI Agents
- **Backend and Tools:** FastAPI, REST APIs, Pydantic, Pytest, SQLite, Docker, Git, GitHub, Jupyter, React, Tailwind CSS, HTML, CSS

Education

B.Tech in Artificial Intelligence and Data Science, Poornima University, Jaipur, Rajasthan · Expected 2028 · CGPA 8.5/10

Certifications

- Python — GeeksforGeeks

- LangChain and LangGraph — IBM
- Claude certificate — Anthropic (replace with exact course title)